

Do Vision Models Truly Forget?

New Findings from Representation-Level Certification of Visual Unlearning in Vertical Federated Learning

Zhenyu Yu^{1,†} Yangchen Zeng^{2,†} Chunlei Meng¹ Guangzhen Yao³ Shuigeng Zhou^{1,*}
¹ Fudan University ² Southeast University ³ Northeast Normal University
[†] Co-first author ^{*} Corresponding author



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Output-level forgetting is an **illusion**. Behaviour is suppressed, yet the forgotten class survives inside the representation, still linearly recoverable.

+15.4 pp
forgetting gap

for Boundary Unlearning on COVID-19, whose output-level $y_u = 0\%$ reads as “forgotten”.

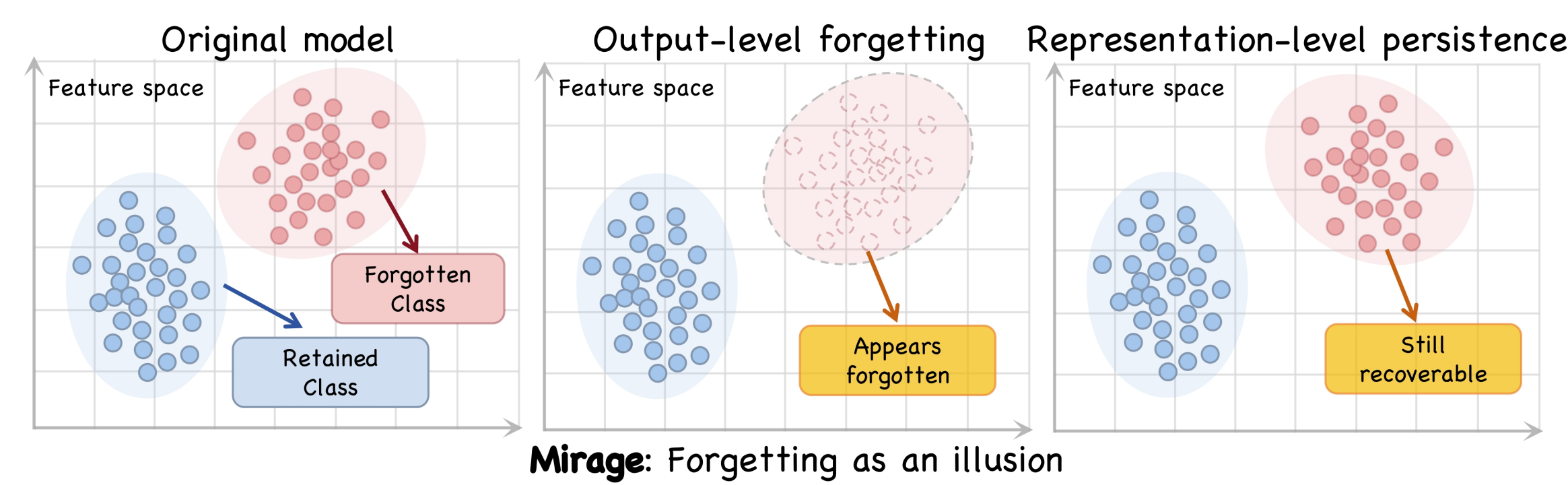
94.7%
still recoverable

linear-probe recovery of the “forgotten” class from frozen features, against 79.2% for Retrain.

≈ 50%
sample-level = chance

LPR sits at chance for every method, Retrain included, so linear probes cannot certify sample-level forgetting.

1 Motivation: The Forgetting Illusion



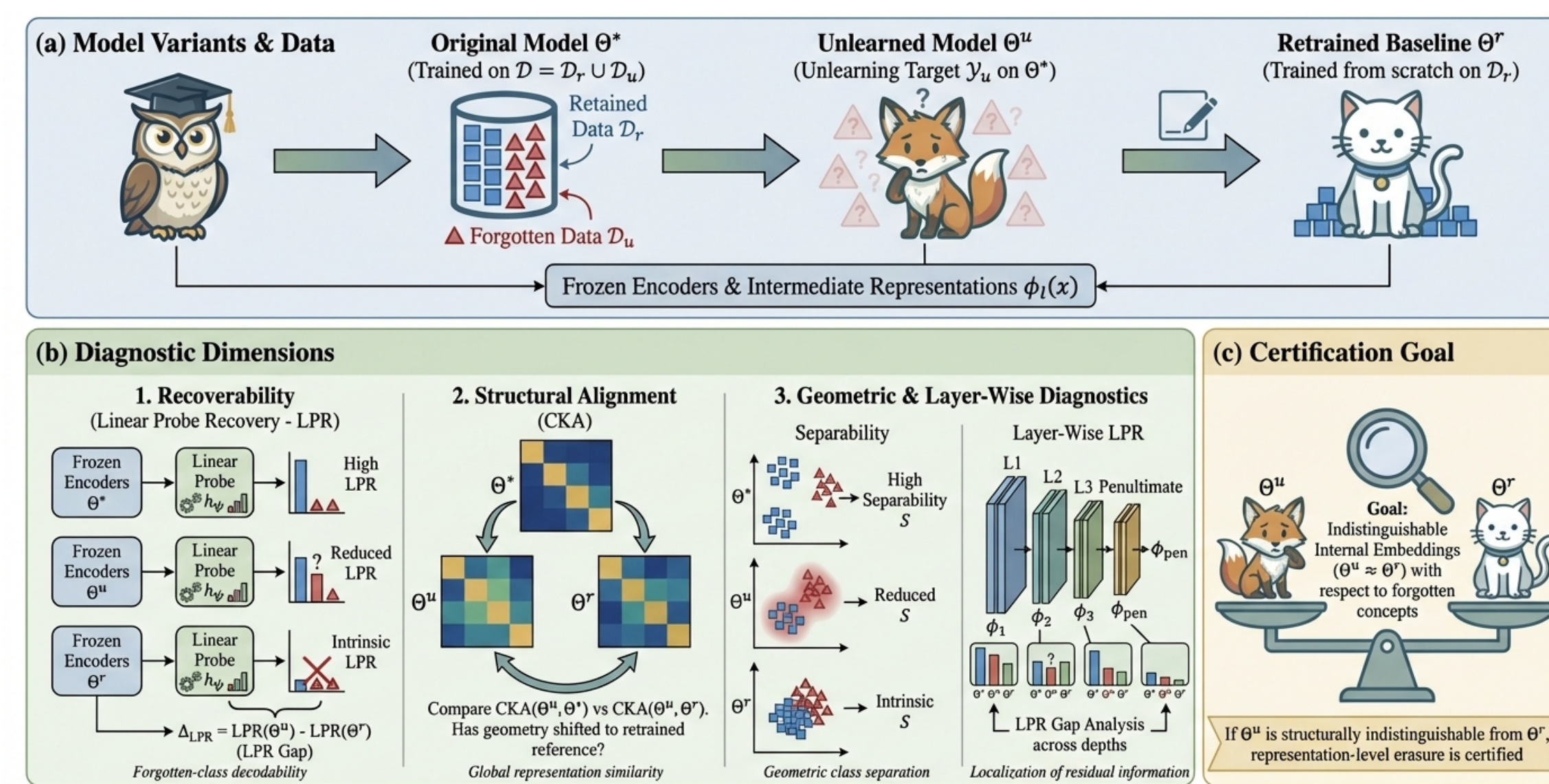
VFL unlearning is certified *only* by output behaviour: the forgotten-label accuracy y_u . But deep networks store class identity as **stable feature geometry**, so a model can drive $y_u \rightarrow 0$ while the class stays linearly decodable in its embeddings. In VFL those embeddings are shared across parties, turning residual class structure into a real **privacy risk**.

2 Experimental Setup

- 7 datasets** MNIST, CIFAR-10, CIFAR-100, ModelNet, Brain Tumor MRI, COVID-19 Radiography, Yahoo Answers.
- 7 unlearning methods** FT, Fisher, Amnesiac, UNSIR, Boundary Unlearning (BU), SSD and Target, every one audited against a from-scratch **Retrain** reference.
- Protocol** ResNet-18 (vision) / MLP (text); VFL with 2 passive parties holding disjoint feature partitions; mean over 3 seeds.
- Probe** l_2 -regularised logistic regression ($C = 1.0$) on frozen penultimate features; linear CKA on 5 000 samples.
- Prior certification** Acc_r (retained accuracy $\uparrow$) and y_u (forgotten-label accuracy $\downarrow$), both output-level only.

Absolute LPR is uninformative. Retrain alone already reaches 54–83%, so the **gap to Retrain** is the principled criterion.

3 Mirage: Representation-Level Auditing



$$\Delta_{\text{LPR}} = \text{LPR}(\Theta^u) - \text{LPR}(\Theta^r)$$

Freeze the unlearned model Θ^u and probe its embeddings against a from-scratch retrained reference Θ^r , which captures the *intrinsic* class geometry of the retained data. A **positive gap** therefore reveals residual, recoverable class structure that genuine forgetting would have erased.

4 Four Diagnostics

- LPR + gap** is the forgotten class still linearly decodable from frozen features?
- CKA** is the geometry aligned with the retrained model, or still with the original?
- Separability** Fisher-inspired separability score F , relative to the retrained baseline.
- Layer-wise** does the residual persist across depth? **BU and FT stay above Retrain** at both the mid and the penultimate layer.

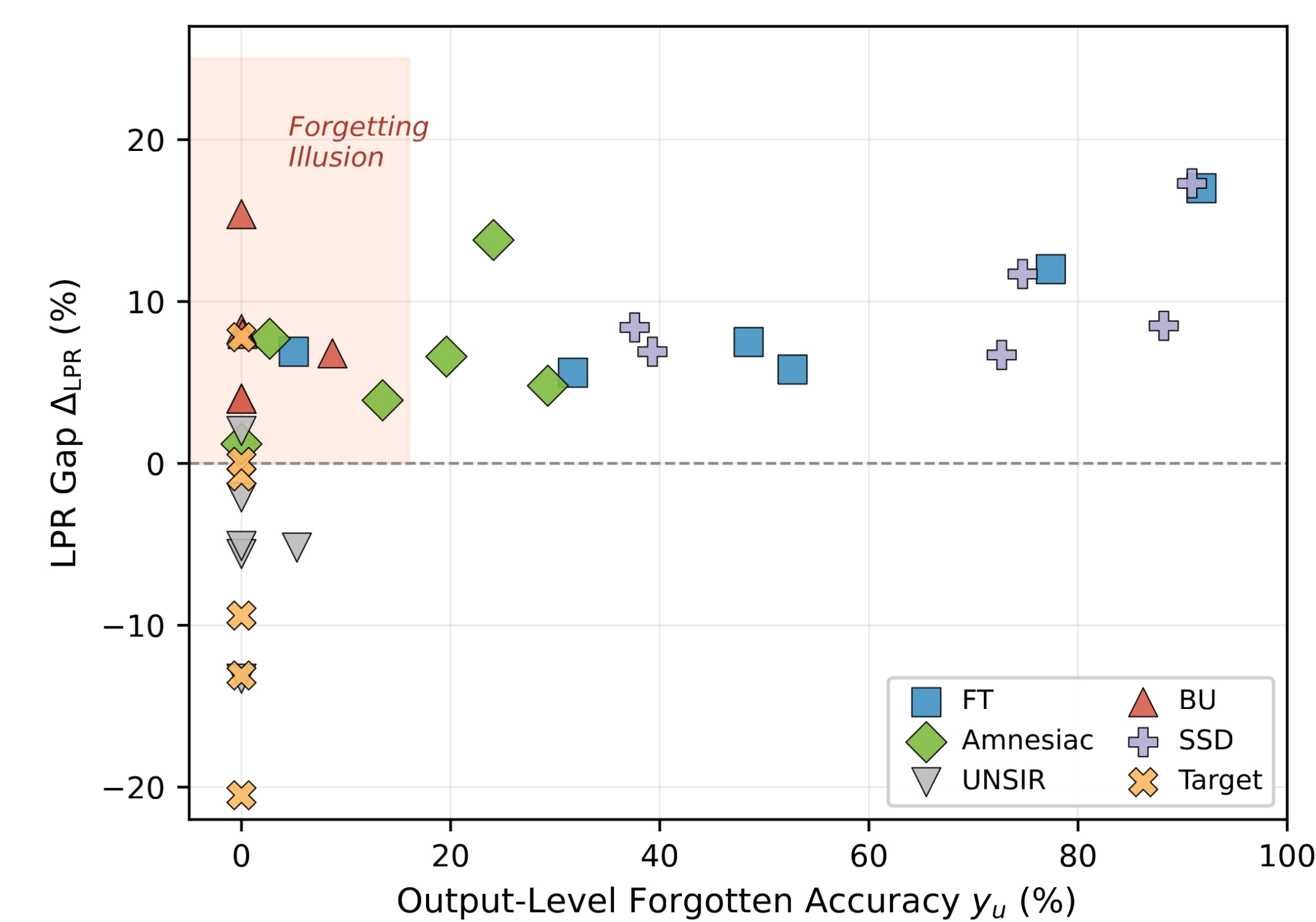
Certified when $|D(\Theta^u) - D(\Theta^r)| \leq \epsilon$ for every diagnostic D .

5 The Forgetting Gap

	CIFAR-10		COVID-19		Brain Tumor	
Method	LPR	Δ	LPR	Δ	LPR	Δ
Retrain	82.8	—	79.2	—	79.8	—
FT	90.4	+7.5	96.2	+17.0	91.8	+12.0
Amnesiac	86.7	+3.9	93.0	+13.8	86.4	+6.6
UNSIR	69.5	−13.3	77.2	−2.1	74.2	−5.6
BU	86.8	+4.0	94.7	+15.4	88.1	+8.3
SSD	91.3	+8.5	96.5	+17.3	91.5	+11.7
Target	82.0	−0.8	69.9	−9.4	79.9	+0.1

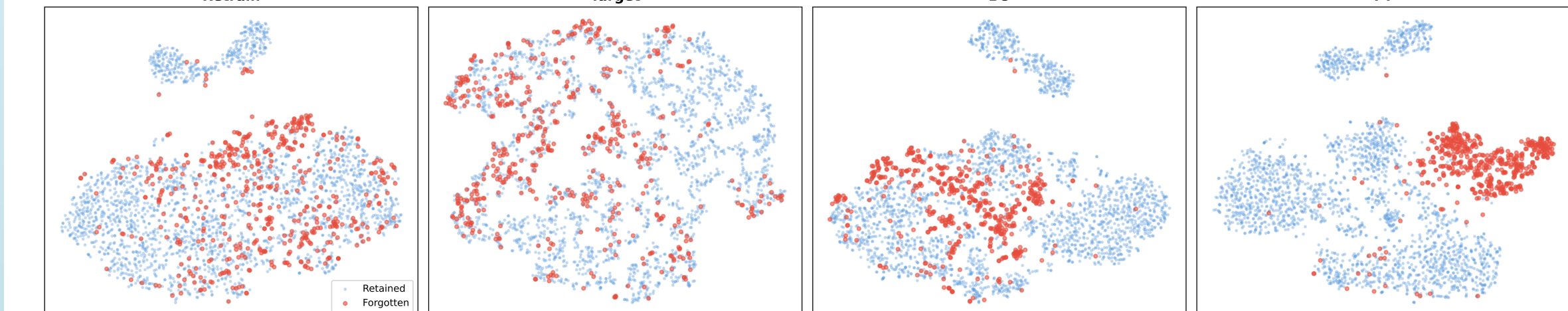
Linear Probe Recovery (%) of the forgotten class; Δ is the gap to Retrain, red where $\Delta > 5$. BU passes output certification on COVID-19 ($y_u = 0$) yet sits **+15.4 pp** above the retrained baseline. Fisher is omitted: it collapses utility on all vision datasets.

6 Two Failure Modes



Boundary Unlearning hides in the **forgetting-illusion** region ($y_u \approx 0$, $\Delta > 0$). **Target** only “forgets” by collapsing the whole feature space on the vision datasets, losing utility with it ($\text{Acc}_r = 34.3\%$ on COVID-19).

7 Geometry Persists



t-SNE of bottom-model features (COVID-19). Under **BU** the forgotten cluster (red) stays intact, almost identical to **Retrain**; **Target** scatters every class.

8 Class-Sample Asymmetry

Class-level forgetting leaves strong traces: LPR reaches ~99% on MNIST and >96% on COVID-19, and the gap persists across depth.

Sample-level forgetting (5% / 10% of random samples) sits at **chance ≈ 50%** for every method, Retrain included.

→ Linear probes are necessary for class-level auditing, but insufficient for sample-level verification.

9 Findings & Takeaway

- Unlearning trilemma.** No evaluated method achieves utility, output forgetting and representation forgetting at once.
- Strength ≠ erasure.** BU’s gap holds at $\approx +15$ pp from 1 to 20 epochs on COVID-19, while CIFAR-10 utility falls from 89.7% (Retrain) to 21.8%.
- Scaling.** FT and BU keep positive gaps as features are split over $K = 2, 4, 8$ passive parties.

Certify unlearning with Δ_{LPR} against a retrained reference. Real forgetting must reshape feature geometry, not just the classifier.